

Notes: Do, Levchenko, and Raddatz (JDE, 2016)

Comparative Advantage, International Trade, and Fertility

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1 Big picture

- **Question.** Does a country's comparative advantage in female-labor-intensive industries reduce fertility by raising female wages and therefore the opportunity cost of children?
- **Setting.** The paper combines a simple two-country trade-and-fertility model with a broad cross-country data set covering 79 tradeable sectors in up to 146 countries over roughly five decades.
- **Core empirical challenge.** Observed export patterns and fertility are jointly determined. A country with lower fertility may have more female labor supply and therefore may also export relatively more in female-intensive sectors. So a simple correlation between export composition and fertility is not causal.
- **Main conceptual contribution.** The paper brings Becker's opportunity-cost-of-time view of fertility into a Ricardian trade setting with sectoral gender intensity. The key idea is that trade patterns affect the demand for female labor, and that labor-demand shift feeds back into fertility.
- **Main theoretical prediction.** If a country has comparative advantage in female-intensive goods, equilibrium female wages are higher and fertility is lower.
- **Main empirical design.** The paper constructs a country-level measure of the *female labor needs of exports* (FLNX), then instruments for it using a geography-based gravity strategy in the spirit of Frankel and Romer (1999) and Do and Levchenko (2007).
- **Main cross-sectional result.** In the preferred 2SLS specification with regional dummies, the elasticity of fertility with respect to FLNX is about -0.59 . A 10% increase in FLNX lowers fertility by about 5.9%.
- **Economic magnitude.** Moving from the 25th to the 75th percentile of FLNX lowers fertility by about 20%, or roughly 0.6 births per woman relative to the sample mean of 3.1.
- **Main panel result.** The negative relationship survives country fixed effects and time fixed effects in 5-year and 10-year panels from 1962 to 2007.
- **Main credibility result.** The estimates are robust to alternative treatments of zero trade flows, dropping major country groups, adding many fertility covariates, and using an alternative multi-country measure of sectoral female intensity.
- **Economic takeaway.** Comparative advantage does not just affect production and wages. It also affects nonmarket outcomes such as fertility by shifting the opportunity cost of women's time.

2 Data and measurement (key objects)

2.1 Why this setting is useful

- The paper needs three ingredients at once: sector-level gender intensity, country-level export composition, and broad fertility data.

- The cross-country setting is essential because comparative advantage is fundamentally about differences across countries in technologies, endowments, and trade patterns.
- The long time span matters because it allows the paper to study both cross-sectional differences and within-country changes in trade structure and fertility.

Interpretation. This is not a micro paper about one country’s labor market. It is a macro-development paper asking whether the gender content of a country’s specialization pattern helps shape demographic outcomes.

2.2 Data sources and sample construction

The empirical analysis combines several data sources:

- bilateral trade flows from COMTRADE, aggregated to a common sector classification;
- sector-level female employment shares from the U.S. Bureau of Labor Statistics (BLS);
- fertility from the World Development Indicators;
- income and openness from the Penn World Tables;
- additional controls such as female education, child labor, infant mortality, inequality, and democracy from standard international sources.

The baseline sector sample contains 79 tradeable sectors: 10 in agriculture and mining and the rest in manufacturing. The cross-sectional regressions use country averages for 2000–2007. The panel specifications use non-overlapping 5-year and 10-year averages from 1962 to 2007.

Interpretation. The paper’s central empirical object is not a household or a firm. It is a country-decade or country-period observation built from sectoral export composition and demographic outcomes.

2.3 Female labor intensity by sector

For each sector i , the paper defines female labor intensity as

$$FL_i = \frac{\text{female employment in sector } i}{\text{total employment in sector } i}.$$

The BLS-based sector ranking is very wide:

- cut-and-sew apparel manufacturing has $FL_i = 0.66$;
- footwear and related products are around 0.56;
- logging is only 0.05 and coal mining only 0.06;
- the mean across sectors is about 0.27.

Interpretation. The cross-sector variation is exactly what the paper needs. Exporting a dollar of apparel is treated as embodying much more female labor demand than exporting a dollar of metals or mining output.

2.4 The female labor needs of exports

The main explanatory variable is the country-level *female labor needs of exports*:

$$FLNX_{ct} = \sum_{i=1}^I \omega_{ict}^X FL_i,$$

where ω_{ict}^X is sector i 's share in country c 's total exports in period t .

- If a country exports mostly apparel, textiles, and other female-intensive goods, $FLNX$ is high.
- If a country exports mostly mining, petroleum, or heavy male-intensive sectors, $FLNX$ is low.

Interpretation. $FLNX$ is the empirical proxy for “comparative advantage in female-intensive sectors.” It is not a direct wage variable, but the paper interprets it as shifting the relative demand for female labor and thus women’s opportunity cost of childbearing.

2.5 Main outcome: fertility

The dependent variable is the fertility rate, measured as births per woman. The baseline regressions use log fertility.

The paper highlights the scale of variation:

- in the 2000s, mean fertility is 1.64 in the OECD sample and 3.34 in the non-OECD sample;
- in the same period, the pooled mean across the analysis sample is about 3.1 births per woman;
- the cross-country standard deviation is about 1.7, and the 10th–90th percentile range runs roughly from 1.3 to 6.1 births per woman.

Interpretation. The variation is economically large enough that even moderate elasticities of fertility with respect to export composition translate into meaningful changes in births per woman.

2.6 Controls

The baseline controls are:

- PPP-adjusted GDP per capita;
- total trade openness;
- regional dummies in the cross section;
- country fixed effects and time fixed effects in the panel.

Additional controls in robustness checks include:

- female educational attainment;
- child labor prevalence;
- infant mortality;

- the Gini coefficient;
- the Polity 2 democracy index.

Interpretation. These controls are designed to separate the effect of the gender content of exports from the many other standard correlates of fertility.

2.7 Descriptive facts

The descriptive tables show several useful patterns:

- OECD countries have relatively stable average $FLNX$ over time, around 0.27–0.28.
- Non-OECD countries have lower average $FLNX$ in the 1960s (0.255) but higher average $FLNX$ by the 2000s (0.291), with much larger dispersion.
- The countries with the highest $FLNX$ in the 2000s are mostly apparel exporters such as Lesotho, Cambodia, Bangladesh, and Haiti.
- The countries with the lowest $FLNX$ are mostly natural-resource exporters such as Mauritania, Gabon, Papua New Guinea, Mozambique, and Iraq.
- Countries with the largest increases in $FLNX$ since the 1960s experienced much larger fertility declines than countries with the largest decreases in $FLNX$.

Interpretation. The raw patterns already line up with the theory: countries specialized in female-intensive export sectors look demographically different from natural-resource exporters.

3 Models and empirical specifications (all equations + interpretation)

3.1 Theoretical model and economic mechanism

The model has two countries, two sectors, and gender-specific labor:

- sector F uses female labor and capital;
- sector M uses male labor and capital;
- countries differ in relative productivity F^c/M^c across these sectors.

Representative-household utility is

$$u(C_F^c, C_M^c, N^c) = (C_F^c)^\eta (C_M^c)^{1-\eta} + v(N^c),$$

with $v(\cdot)$ increasing and concave.

Production in sector $i \in \{F, M\}$ is

$$Y_i^c(K_i, L_i) = \theta_i^c K_i^\alpha L_i^{1-\alpha}.$$

Women split time between market work and childrearing, so female market labor supply is

$$L_F^c = 1 - \lambda N^c.$$

Interpretation. The model deliberately gives women a larger childrearing burden. That is what makes a rise in female wages translate into a rise in the opportunity cost of fertility.

3.2 Production-trade equilibrium and fertility equilibrium

For a given fertility level, female labor supply determines each country's effective female labor endowment. Comparative advantage in the female-intensive sector then raises the relative size of that sector and raises female wages.

The labor-supply side is upward sloping because the utility function is additively separable in fertility and consumption, so the substitution effect dominates: higher female wages reduce fertility. The labor-demand side is downward sloping because more female labor supply expands the female-intensive sector and changes the relative factor-proportions channel.

Interpretation. The model boils down to an intersection of female labor supply and labor demand. Trade structure shifts labor demand; fertility adjusts through labor supply.

3.3 Main theorem

The paper's central theoretical result is:

Theorem 1 (cross-sectional comparison). If country c has Ricardian comparative advantage in the female-labor-intensive sector,

$$\frac{F^c}{M^c} > \frac{F^{-c}}{M^{-c}},$$

then equilibrium fertility is lower in country c .

Interpretation. Comparative advantage in female-intensive sectors and lower fertility are not just correlated in the model. They are equilibrium counterparts of the same force: stronger demand for female labor.

3.4 Baseline cross-sectional specification

The main cross-sectional regression is

$$N_c = \alpha + \beta FLNX_c + \gamma Z_c + \varepsilon_c,$$

where N_c is fertility and Z_c is the vector of controls.

The key prediction is

$$\beta < 0.$$

Interpretation. A country with more female-intensive exports should have lower fertility, conditional on income, openness, and region.

3.5 Panel specification

The panel regression is

$$N_{ct} = \alpha + \beta FLN X_{ct} + \gamma Z_{ct} + \delta_c + \delta_t + \varepsilon_{ct},$$

where δ_c are country fixed effects and δ_t are time effects.

- The cross-sectional 2SLS is the paper's sharpest causal design.
- The panel cannot use the same gravity IV with country fixed effects, but it controls for time-invariant country heterogeneity directly.

Interpretation. The two strategies complement one another. The cross section is stronger on instrumentation; the panel is stronger on absorbing fixed country traits.

3.6 Gravity instrument for trade patterns

The gravity equation is estimated separately by sector:

$$\log X_{icd} = \alpha_i + \eta'_i G_{cd} + \varepsilon_{icd},$$

where G_{cd} collects bilateral geographic variables such as distance, border, landlocked status, population, and area.

The crucial insight is that the gravity coefficients η_i differ across sectors. That implies that geography predicts different trade patterns across sectors, not just different total trade volumes.

Predicted exports are aggregated across destinations:

$$\hat{X}_{ic} = \sum_{d \neq c} \exp(\widehat{\log X_{icd}}),$$

and predicted export shares are then used to construct the instrumented female labor needs of exports:

$$\widehat{FLN X}_c = \sum_{i=1}^I \hat{\omega}_{ic}^X FL_i.$$

Interpretation. The IV is built from exogenous geography plus sector-specific gravity sensitivities. That is what allows the paper to instrument for the *composition* of exports instead of total trade alone.

3.7 Why the first stage works

The paper gives two arguments.

- **Economic logic.** Heavy, bulky, or differentiated goods respond differently to distance and borders than light or standardized goods.
- **Data evidence.** The estimated distance coefficient varies widely across sectors, from about -2.42 to -0.10 . The border coefficient also varies enormously across sectors.

Interpretation. If all sectors responded to geography in exactly the same way, the gravity approach could only instrument for openness. The whole design depends on cross-sector heterogeneity in gravity coefficients.

3.8 Estimation details

- Cross-sectional regressions use 2000–2007 averages.
- Panel regressions use 5-year and 10-year averages to smooth out business-cycle noise.
- Cross-sectional standard errors are robust.
- Panel standard errors are clustered at the country level.

Interpretation. The paper is intentionally conservative about time-series noise and autocorrelation, especially in the long panel.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline estimate

Identification in the cross section comes from country differences in *predicted* female-labor-intensive export patterns generated by geography and sector-specific gravity responses. The core identifying assumption is that, conditional on controls, this geography-driven component of export composition affects fertility only through comparative advantage and its effect on female labor demand.

Interpretation. The paper does not instrument female wages directly. Instead, it instruments a trade-pattern-based proxy for the opportunity cost of women’s time.

4.2 Why reverse causality is a problem in raw data

Raw $FLNX$ is potentially endogenous for at least two reasons:

- lower fertility raises female labor supply, which can itself push specialization toward female-intensive sectors;
- omitted country characteristics could jointly affect fertility and trade patterns.

Interpretation. The IV is not a minor refinement. It is necessary because the theory itself implies joint determination of fertility and specialization.

4.3 Why the geography instrument is plausible

The instrument uses exogenous features such as bilateral distance, common border, country population, area, and landlocked status. These variables are standard gravity shifters and are not chosen in response to modern fertility decisions.

The crucial extra step is that the instrument translates these country characteristics into a predicted *sectoral* export composition using coefficients estimated sector by sector.

Interpretation. A country's geography does not just predict how much it trades. Because sectors differ in how sensitive they are to distance and borders, geography also predicts *what* the country is likely to export.

4.4 Why the paper does not just use trade openness

A rise in trade openness alone is not enough for the paper's question. Two countries can be equally open but specialize in very different sectors. The paper wants to know whether openness *in female-intensive sectors* matters for fertility.

Empirically, the coefficient on total trade openness is near zero in the main cross-sectional specifications, while FLNX is negative and significant.

Interpretation. The result is about comparative advantage, not globalization in the generic sense.

4.5 Zero-trade observations and first-stage robustness

A possible concern is that the log-linear gravity baseline ignores bilateral zero trade flows. The paper addresses this in two ways:

- predicting out-of-sample trade for missing and zero observations using the log-linear gravity model;
- estimating the gravity model in levels with Poisson pseudo-maximum likelihood.

The second-stage results remain very similar under both alternatives.

Interpretation. The IV is not fragile to the treatment of zero trade flows.

4.6 Country-specific gravity variables and exclusion concerns

The instrument uses country population and area, which may seem uncomfortable if they affect fertility directly. The paper therefore does two further checks:

- it rebuilds the instrument excluding exporter population;
- it controls directly for log area and log population in the fertility equation.

The FLNX coefficient remains negative and significant in both cases, though somewhat smaller when exporter population is removed from the instrument.

Interpretation. The identifying power is not coming from a single obviously problematic variable.

4.7 Cross-sectional IV versus panel fixed effects

The paper cannot implement the same gravity IV in a country fixed-effects panel because that would require strong differential time variation in sector-specific gravity coefficients, and the data do not provide enough of that variation.

Interpretation. This is why the paper presents two empirical strategies rather than a single all-purpose design.

4.8 What the paper can and cannot distinguish

What it identifies well.

- Comparative advantage in female-intensive sectors is associated with lower fertility.
- The effect is economically meaningful and robust.

What it cannot identify directly.

- It cannot directly estimate the gender wage gap because broad international wage-gap data are poor and confounded by selection and composition.
- It cannot fully separate whether the effect works purely through wages or also through broader changes in women's labor-market opportunities and bargaining power.

Interpretation. The paper is strongest on the reduced-form relationship predicted by the model. It is more indirect on the exact labor-market channel because the necessary wage data do not exist at comparable quality.

4.9 Relation to the literature

The paper sits at the intersection of several literatures:

- Becker-style fertility and female opportunity cost of time;
- Ricardian trade and comparative advantage;
- globalization and gender outcomes;
- trade and fertility in development economics.

Its distinctive contribution is to show theoretically and empirically that the gender content of comparative advantage itself can shape fertility.

5 Tables and figures

5.1 Table 1: ranking sectors by female labor intensity

Read:

- Table 1 lists sectors from the most female-intensive to the least female-intensive.
- Apparel and textiles sit at the top, while mining and logging sit at the bottom.
- The spread is very large: roughly 0.66 at the top and 0.05 at the bottom.

Economic message. The sector ranking gives the empirical content to the model's "female-intensive" and "male-intensive" goods.

Table 1

Share of female workers in total employment, highest to lowest.

Source: BLS.

Sector name	FL_i
Cut and sew apparel manufacturing	0.66
Footwear; leather; textile and apparel, n.e.c.	0.56
Textile product mills, except carpet and rug	0.55
Soap, cleaning compound, and cosmetics manufacturing	0.51
Sugar and confectionery products	0.48
Pharmaceutical and medicine manufacturing	0.46
Medical equipment and supplies manufacturing	0.44
Fabric mills, except knitting mills	0.43
Carpet and rug mills	0.41
Seafood and other miscellaneous foods, n.e.c.	0.40
Miscellaneous manufacturing, n.e.c.	0.39
Not specified food industries	0.38
Fruit and vegetable preserving and specialty food manufacturing	0.37
Sporting and athletic goods, and doll, toy and game manufacturing	0.37
Not specified manufacturing industries	0.37
Bakeries, except retail	0.37
Animal slaughtering and processing	0.36
Household appliance manufacturing	0.36
Printing and related support activities	0.35
Electronic component and product manufacturing, n.e.c.	0.35
Agricultural chemical manufacturing	0.35
Miscellaneous paper and pulp products	0.35
Navigational, measuring, electromedical, and control instruments manufacturing	0.32
Communications, and audio and video equipment manufacturing	0.32
Rubber product, except tire, manufacturing	0.32
Forestry, except logging	0.31
Electrical lighting and electrical equipment and other electrical component manufacturing, n.e.c.	0.31
Computer and peripheral equipment manufacturing	0.30
Plastics product manufacturing	0.30
Cutlery and hand tool manufacturing	0.29
Commercial and service industry machinery manufacturing	0.28
Resin, synthetic rubber and fibers, and filaments manufacturing	0.28
Glass and glass product manufacturing	0.28
Tobacco manufacturing	0.27
Furniture and fixtures manufacturing	0.27
Dairy product manufacturing	0.26
Beverage manufacturing	0.26
Animal food, grain, and oilseed milling	0.26
Paperboard containers and boxes	0.25
Animal production	0.25
Miscellaneous fabricated metal products manufacturing	0.25
Motor vehicles and motor vehicle equipment manufacturing	0.25
Railroad rolling stock; Other transportation equipment	0.24
Crop production	0.23
Paint, coating, and adhesive manufacturing	0.23
Aircraft and parts manufacturing	0.23
Miscellaneous wood products	0.22
Machinery manufacturing, n.e.c.	0.22
Veneer, plywood, and engineered wood products	0.22
Not specified machinery manufacturing	0.22
Electric power generation, transmission, and distribution	0.22
Industrial and miscellaneous chemicals	0.22
Engines, turbines, and power transmission equipment manufacturing	0.21
Agricultural implement manufacturing	0.21
Miscellaneous petroleum and coal products	0.21
Nonferrous metal (except aluminum) production and processing	0.20
Petroleum refining	0.20
Oil and gas extraction	0.20
Pulp, paper, and paperboard mills	0.19
Pottery, ceramics, structural clay, and plumbing fixtures	0.19
Ordnance; Not specified metal industries	0.18
Prefabricated wood buildings and mobile homes	0.17
Ship and boat building	0.17
Structural metals, and boiler, tank, and shipping container manufacturing	0.17
Miscellaneous nonmetallic mineral product manufacturing	0.17

Table 1 (continued)

Sector name	FL_i
Metal ore and nonspecified type of mining	0.13
Nonmetallic mineral mining and quarrying	0.11
Sawmills and wood preservation	0.11
Cement, concrete, lime, and gypsum product manufacturing	0.10
Coal mining	0.06
Logging	0.05
Mean	0.27

Notes: this table reports the share of female workers in total employment by sector in the US.

Figure 1: Table 1 from the paper: female employment shares by sector.

5.2 Tables 2–4: descriptive patterns in FLNX and fertility

Read:

- Table 2 shows that OECD countries have lower fertility and lower dispersion in fertility than non-OECD countries, but non-OECD countries' average FLNX rises strongly over time.
- Table 3 shows that the highest-FLNX countries are mainly apparel exporters, while the lowest-FLNX countries are resource exporters.
- Table 4 shows that countries with the largest increases in FLNX since the 1960s also tend to have much larger fertility declines.

Economic message. Before any formal regression, the descriptive evidence already looks consistent with the paper's story.

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Table 2

Summary statistics for female labor need of exports and fertility.

OECD				Non-OECD			
	Mean	St. dev.	Countries	Mean	St. dev.	Countries	
<i>Panel A: female labor need of exports</i>							
1960s	0.272	0.036	21	0.255	0.055	100	
1970s	0.274	0.034	21	0.260	0.061	101	
1980s	0.273	0.037	21	0.265	0.068	101	
1990s	0.280	0.036	21	0.288	0.080	121	
2000s	0.279	0.030	21	0.291	0.094	125	
<i>Panel B: fertility rates</i>							
1960s	2.784	0.452	21	6.165	1.380	100	
1970s	2.120	0.449	21	5.748	1.607	101	
1980s	1.735	0.257	21	5.109	1.768	101	
1990s	1.634	0.249	21	3.951	1.829	121	
2000s	1.641	0.255	21	3.341	1.664	125	

Table 4

FLNX: Top 10 and bottom 10 changers since 1960s.

Largest increase in FLNX		Largest decrease in FLNX	
Cambodia	0.342	Sudan	−0.040
Haiti	0.298	Uruguay	−0.044
Honduras	0.252	Papua New Guinea	−0.056
El Salvador	0.221	Central African Republic	−0.058
Albania	0.209	Australia	−0.060
Tunisia	0.192	Angola	−0.063
Nicaragua	0.165	Hong Kong, China	−0.064
Jordan	0.161	Mozambique	−0.102
Morocco	0.158	Tanzania	−0.109
Guatemala	0.141	Cuba	−0.113

Notes: This table reports the 10 countries with the largest increases and the largest decreases in FLNX. Change is calculated as the difference between the FLNX in the 2000s and that in the 1960s.

Highest FLNX		Lowest FLNX	
Lesotho	0.612	Mauritania	0.152
Cambodia	0.609	Gabon	0.175
Bangladesh	0.595	Papua New Guinea	0.181
Haiti	0.593	Mozambique	0.184
Mauritius	0.512	Kazakhstan	0.189
El Salvador	0.490	Guinea	0.190
Sri Lanka	0.485	Congo, Rep.	0.193
Honduras	0.482	Venezuela, RB	0.196
Pakistan	0.474	Libya	0.198
Madagascar	0.440	Iraq	0.198

Notes: This table reports the 10 countries with the highest, and 10 countries with the lowest FLNX.

Table 4
FLNX: Top 10 and bottom 10 changers since 1960s.

Largest increase in FLNX		Largest decrease in FLNX	
Cambodia	0.342	Sudan	−0.040
Haiti	0.298	Uruguay	−0.044
Honduras	0.252	Papua New Guinea	−0.056
El Salvador	0.221	Central African Republic	−0.058
Albania	0.209	Australia	−0.060
Tunisia	0.192	Angola	−0.063
Nicaragua	0.165	Hong Kong, China	−0.064
Jordan	0.161	Mozambique	−0.102
Morocco	0.158	Tanzania	−0.109
Guatemala	0.141	Cuba	−0.113

Notes: This table reports the 10 countries with the largest increases and the largest decreases in FLNX. Change is calculated as the difference between the FLNX in the 2000s and that in the 1960s.

Figure 2: Tables 2–4 from the paper: summary statistics, top and bottom FLNX countries, and largest FLNX changers.

5.3 Table 5 and Figures 1–2: main cross-sectional evidence

Read:

- In OLS, the coefficient on log FLNX is between about -0.24 and -0.34 .
- In 2SLS, it becomes much larger in absolute value: about -0.73 without regional dummies and about -0.59 with regional dummies.
- The first stage is strong, with excluded-instrument F-statistics around 44–56 in the main columns.
- Figure 1 shows a clear negative partial correlation between fertility and FLNX.
- Figure 2 shows a clear positive first-stage relationship between actual FLNX and predicted FLNX.

Economic message. This is the heart of the paper. Countries whose export bundles are more female-intensive have lower fertility, and the geography-based first stage is strong enough to take the causal interpretation seriously.

Table 5
Cross-sectional results, 2000–2007.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	OLS	OLS	2SLS	2SLS	2SLS	2SLS	2SLS	2SLS
Dependent variable: (log) fertility rate								
(Log) female labor	–0.34*** (0.08)	–0.34*** (0.08)	–0.73*** (0.18)	–0.59*** (0.15)	–0.65*** (0.19)	–0.60*** (0.18)	–0.41*** (0.14)	–0.46*** (0.18)
Need of exports	–0.04 (0.04)	0.01 (0.03)	–0.02 (0.04)	0.03 (0.03)	0.03 (0.03)	0.03 (0.03)	0.02 (0.03)	0.02 (0.03)
(Log) GDP per capita	–0.35*** (0.02)	–0.22*** (0.03)	–0.35*** (0.02)	–0.23*** (0.02)	–0.23*** (0.03)	–0.23*** (0.03)	–0.22*** (0.02)	–0.22*** (0.02)
Log (area)								0.01 (0.02)
Log (population)								–0.02 (0.02)
Constant	5.34*** (0.32)	3.62*** (0.42)	6.57*** (0.59)	5.09*** (0.59)	5.31*** (0.75)	5.14*** (0.73)	4.42*** (0.55)	4.62*** (0.89)
R ²	0.678	0.831	First stage					
Dependent var. (log) FLNX			1.84*** (0.24)	1.68*** (0.25)				3.00*** (0.66)
(Log) predicted FLNX					1.25*** (0.25)			
(Log) predicted FLNX (Out of sample)						0.96*** (0.18)		
(Log) predicted FLNX (Poisson)							4.30*** (0.61)	
(Log) predicted FLNX (No population)								
F-test			56.42	44.01	24.16	27.00	49.81	20.39
First stage partial R ²			0.290	0.277	0.165	0.193	0.307	0.174
Region dummies	No	Yes	No	Yes	Yes	Yes	Yes	Yes
Observations	146	146	146	146	146	146	146	146

Notes: Robust standard errors in parentheses; * significant at 10%; ** significant at 5%; *** significant at 1%. All variables are averages over the period 2000–2007 and in natural logs. Variable definitions and sources are described in detail in the text.

Figure 3: Table 5 from the paper: main cross-sectional OLS and 2SLS results.



Partial correlation between fertility and FLNX. Notes: this figure displays the partial correlation between FLNX and fertility, in logs, after controlling for openness, per capita income, and regional dummies (see Column 2 of Table 5).

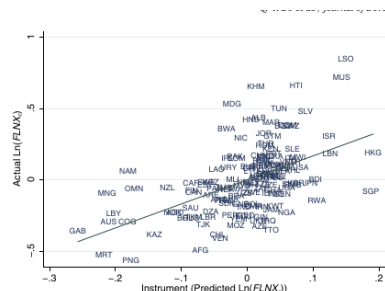


Fig. 2. First stage: partial correlation between FLNX and FLNX. Notes: this figure presents the partial correlation plot from the first stage regression between the actual value of FLNX and the instrument.

Figure 4: Figures 1 and 2 from the paper: partial correlation and first-stage plot.

5.4 Table 6: panel evidence

Read:

- The coefficient on log FLNX is negative and significant in pooled panels, country fixed-effects panels, and country-plus-time fixed-effects panels.
- The estimates are sizable for both 5-year and 10-year averages.
- Adding female educational attainment leaves the sign intact and the coefficient still statistically meaningful.

Economic message. The cross-sectional IV result is not being driven only by permanent differences across countries. Within-country shifts in trade composition toward female-intensive exports are also associated with lower fertility.

Table 6
Panel Results, 1962–2007.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Five-year averages				Ten-year averages			
<i>Dependent variable: (log) fertility rate</i>								
(Log) female labor	−0.60*** (0.09)	−0.54*** (0.10)	−0.24*** (0.09)	−0.20** (0.09)	−0.61*** (0.10)	−0.58*** (0.12)	−0.27*** (0.10)	−0.23** (0.11)
Need of exports	−0.01 (0.03)	−0.15*** (0.04)	−0.01 (0.03)	0.00 (0.03)	−0.01 (0.03)	−0.15*** (0.05)	−0.01 (0.04)	0.01 (0.04)
(Log) openness	−0.35*** (0.02)	−0.35*** (0.05)	−0.19*** (0.04)	−0.18*** (0.05)	−0.35*** (0.02)	−0.37*** (0.06)	−0.20*** (0.05)	−0.20*** (0.05)
(Log) GDP per capita				0.01 (0.04)				0.00 (0.04)
(Log) female Educational attainment	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Country FE	No	No	Yes	Yes	No	No	Yes	Yes
Year FE	0.594	0.889	0.936	0.934	0.602	0.899	0.942	0.940
R ²	1254	1254	1254	1109	630	630	630	557
Observations								

Notes: Standard errors clustered at the country level in parentheses; * significant at 10%; ** significant at 5%; *** significant at 1%. All of the variables are 5-year averages (left panel) or 10-year averages (right panel) over the time periods spanning 1962–2007, and in natural logs. Variable definitions and sources are described in detail in the text.

Figure 5: Table 6 from the paper: panel results using 5-year and 10-year averages.

5.5 Table 7: alternative specifications and added controls

Read:

- The result survives when the key regressor is female labor needs of *net* exports rather than gross exports.
- It also survives when the regressions add female schooling, child labor, infant mortality, inequality, and democracy controls one at a time.
- In every case, the FLNX coefficient remains negative and statistically significant.

Economic message. The FLNX result is not just proxying for one standard development covariate omitted from the fertility regression.

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Table 7
Alternative specifications and controls: cross-sectional 2SLS results, 2000–2007.

	(1)	(2)	(3)	(4)	(5)	(6)
<i>Dependent variable: (log) fertility rate</i>						
(Log) FLNX		−0.51*** (0.13)	−0.58*** (0.14)	−0.42*** (0.15)	−0.37*** (0.14)	−0.54*** (0.14)
FLNX − FLNI	−0.02*** (0.01)					
(Log) openness	0.05 (0.04)	0.04 (0.05)	0.04 (0.05)	0.02 (0.03)	−0.03 (0.05)	0.04 (0.03)
(Log) GDP per capita	−0.24*** (0.03)	−0.20*** (0.03)	−0.23*** (0.04)	−0.14*** (0.04)	−0.24*** (0.03)	−0.22*** (0.03)
(Log) female Educational attainment		−0.13** (0.06)				
Child labor indicator			0.01*** (0.00)			
(Log) infant mortality				0.12*** (0.04)		
Gini coeff					0.62** (0.30)	
Polity 2 indicator						0.00 (0.00)
Constant	3.31*** (0.28)	4.55*** (0.61)	4.85*** (0.78)	3.18*** (0.76)	4.34*** (0.60)	4.53*** (0.59)
(Log) predicted FLNX		1.77*** (0.26)	1.69*** (0.33)	1.70*** (0.30)	1.77*** (0.33)	1.68*** (0.27)
Predicted FLNX	1.62*** (0.31)					
F-test	26.59	48.14	25.92	31.81	26.38	38.88
First stage partial R ²	0.205	0.319	0.275	0.254	0.291	0.271
Region dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	146	126	103	145	103	145

Notes: Robust standard errors in parentheses; * significant at 10%; ** significant at 5%; *** significant at 1%. All variables are averages over the period 2000–2007. Variable definitions and sources are described in detail in the text.

Figure 6: Table 7 from the paper: alternative specifications and additional controls.

5.6 Tables 8 and 9: robustness across subsamples and decades

Read:

- Table 8 shows that the result survives after dropping outliers, dropping OECD countries, dropping Sub-Saharan Africa, dropping the Middle East and North Africa, and dropping mining exporters.
- Table 9 shows that the effect is present in the 1970s, 1980s, and 1990s and is absent only in the 1960s, when global trade integration was much weaker.

Economic message. The result is not driven by one region, one set of exporters, or one specific decade.

5.7 Appendix evidence on female labor force participation

Read:

- Appendix B asks whether the same trade pattern that lowers fertility also raises female labor force participation (FLFP).
- The evidence is supportive but more conditional: FLNX tends to raise FLFP mainly at lower levels of income and female educational attainment.

Economic message. The FLFP results point in the same qualitative direction as the model, but fertility is the cleaner and stronger empirical margin in this paper.

6 Short summary

- The paper asks whether comparative advantage in female-intensive sectors lowers fertility by raising women's opportunity cost of time.
- It builds a simple trade model with sector-specific gender intensity and endogenous fertility, then takes the prediction to the data.
- The key empirical variable is the female labor needs of exports, a country-level average of sectoral female employment shares weighted by export composition.
- Because export composition and fertility are jointly determined, the paper instruments FLNX using geography and sector-specific gravity coefficients.
- The preferred cross-sectional 2SLS estimates imply that more female-intensive export baskets are associated with substantially lower fertility.
- Panel regressions with country and time fixed effects tell the same broad story.
- The result is robust to many alternative controls, subsamples, and ways of constructing the instrument.
- The broader message is that trade affects demographic behavior through gendered labor demand.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that countries with comparative advantage in female-intensive sectors have lower fertility, and that this relationship survives a serious attempt at causal identification using geography-based instruments for trade patterns.

7.2 Economic intuition

Trade changes the relative demand for female labor. When a country's export specialization loads more heavily on sectors that employ women, female wages rise relative to male wages, the opportunity cost of children rises, and fertility falls. In the model, fertility choices then reinforce the original comparative advantage.

7.3 Takeaway

The main lesson is that globalization and comparative advantage can shape outcomes that are usually studied outside trade theory. Fertility is one of them. The paper therefore provides a clean bridge between international trade, gendered labor demand, and demographic behavior.